

William Ng

Software Development Engineer

williamng0512@gmail.com • +6011 - 3385 8659 • Penang • [LinkedIn](#) • [Portfolio](#)

SKILLS

Python • Typescript • JavaScript • React • React Native • Vue • SQL • NoSQL • Docker • Git • Firebase • HTML • CSS •
Linux system administration • Raspberry Pi • Microsoft Power Apps • AWS S3 • Microsoft Azure CLI • GitHub Copilot

EXPERIENCE

Software Development Engineer

Nov 2025 – Present

Centific Global Solutions, Remote

- Develop, support, and triage issues for Microsoft Azure CLI, covering feature development, bug fixes, and resolution of customer-reported issues.
- Drive migration work across Azure CLI components to keep the tooling current, reliable, and maintainable.
- Leverage GitHub Copilot to build automation and internal tooling that streamlines development and support workflows.

Programmer Analysts 1

July 2022 – Nov 2025

Jabil Sdn Bhd, Penang

- Developed applications using TypeScript (frontend) and Python (backend), and containerized the system with Docker to streamline deployment and scalability.
- Maintained and optimized application performance, reducing processing times and resolving bugs to ensure smooth operations.
- Collaborated with cross-functional teams to deploy solutions across multiple production areas, increasing operational efficiency and minimizing downtime.
- Provided online user support, assisting operators in troubleshooting issues and ensuring system usability.

Intern - IOT Developer

May 2022 – July 2022

Jabil Sdn Bhd, Penang

- Developed a proof-of-concept (PoC) application demonstrating the feasibility of OCR integration in IoT solutions.

Intern - Web Developer

Feb 2020 – May 2020

Coblix, Penang

- Built a React.js web application seamlessly integrated with Firebase (NoSQL), powering Me Book Asia's platform with modern, scalable architecture.
- Engineered data migration from traditional SQL to flexible NoSQL databases, unlocking improved performance and adaptability.
- Collaborated on React Native mobile app enhancements, elevating user engagement through polished UI and smoother performance.

PROJECTS

Resume Builder

github.com/william051200/i-build-resume

Built a browser-based resume builder (Vue 3, Vite, TypeScript) that generates clean, print-ready resumes entirely client-side, with JSON import/export and a live preview. Designed a structured, type-safe data model for experience, education, projects, skills, and certifications, rendered into an ATS-friendly, print-optimized layout.

GitHerd

github.com/william051200/githerd

Built a Windows CLI tool that syncs many Git repositories in parallel from a single command, with live per-repo progress bars. Implemented configurable per-repo workflows (fetch + fast-forward-merge + push, or plain pull) driven by a JSON config, with an optional GUI for editing. Added an auto-update mechanism via GitHub Releases with in-place updates and config backups, plus a one-line PowerShell installer.

ArtifactLens

github.com/william051200/ArtifactLens

Built a Python desktop GUI to search Azure DevOps Artifacts feeds by package version, integrating the Azure DevOps REST API with PAT-based authentication. Packaged the app as a dependency-free Windows executable (PyInstaller) with a one-line PowerShell installer that creates Start Menu and Desktop shortcuts. Implemented an auto-update system that checks GitHub Releases and installs new versions in place.

In-House Smart Camera Inspection System

Developed an initial prototype that was enhanced into a full-fledged proof-of-concept application with AI models for real-time quality checks. Designed and implemented RESTful APIs to enable robust communication between the user interface and backend services. Architected and developed Identity and Access Management (IAM) and Single Sign-On (SSO) solutions using JWT, aligning with corporate security standards. Integrated GPIO and MQTT communication channels to support real-time hardware interaction and messaging.

OCR Docker Application

Developed a Python-based prototype containerized with Docker, optimized to run efficiently on Raspberry Pi hardware. Implemented real-time inspection using the Raspberry Pi camera module and USB capture card integration via OpenCV. Integrated AWS S3 for secure and scalable cloud-based data storage. Integrated into the In-House Smart Camera Inspection System, significantly enhancing inspection accuracy and flexibility.

EDUCATION

Bachelor of Information Technology (Honours) in Software Systems Development

2020 – 2022

Tunku Abdul Rahman University College, KL, Malaysia

- CGPA: 3.7242 (Dean's/President's List multiple semesters)
- Developed a mobile application using Kotlin; explored IoT (Raspberry Pi) and fundamentals of blockchain.

Diploma in Science (Information System Engineering)

2018 – 2020

Tunku Abdul Rahman University College, Penang, Malaysia

- CGPA: 3.2976
- Gained strong fundamentals in OOP, networking, and application development.